

Edge AI-Powered Smart Swine Health & Behavior Monitoring System

System Architectural Design & Technical Specification Document

Lead Developer: Le Ba Duc | High School for Gifted Students (HSGS - VNU) | Edge AI, IoT & Computer Vision

1. Project Overview & Clinical Objectives

This engineering project focuses on designing and implementing an autonomous barn surveillance system combining **Edge AI computer vision** with multi-sensor environmental telemetry. The primary clinical objective is early detection of livestock health disorders and behavioral anomalies before visible acute disease progression or herd contagion:

- **Prolonged Recumbency & Inactivity:** Flagging pigs lying motionless beyond critical thresholds (>30 mins), alerting to hypothermia, acute infection, or mortality risk.
- **Locomotion Deficits & Anorexia:** Quantifying diminished pen mobility and feeder visit frequency to catch early-stage lethargy or lameness.
- **Agonistic & Aberrant Behaviors:** Flagging aggressive tail-biting, violent clustering, or erratic movements for immediate operator intervention.

The **AI Camera** serves as the primary non-invasive sensor, performing continuous Day/Night monitoring with infrared illumination. Additionally, integrated microclimate sensors (**DHT22** for temperature/humidity, **MQ-135** for toxic gases NH_3 and CO_2) provide contextual environmental validation to correlate behavioral lethargy with air quality fluctuations.

2. System Interaction Workflow & Sequence Diagram

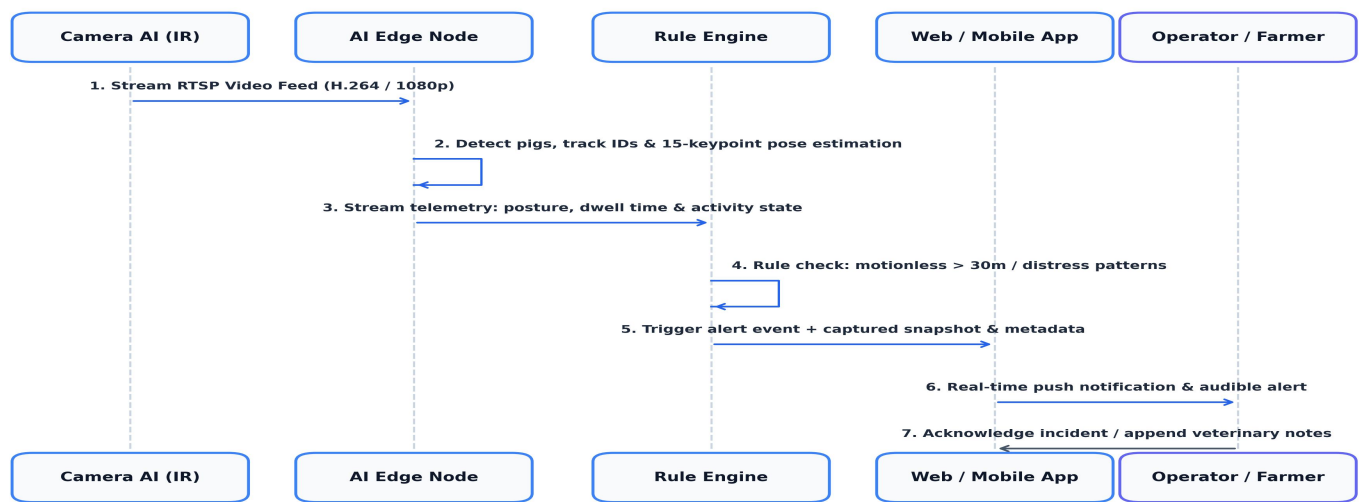


Figure 1: End-to-end Sequence Pipeline (Edge Computer Vision → Rule Engine → Cloud Dashboard → Operator Alert)

Operational Sequence & Alert Dispatch:

- 1. Video Ingestion:** Wide-angle IR camera streams real-time RTSP feed (1080p @ 20 FPS) over local LAN to the Edge AI node.
- 2. Edge Inference:** YOLOv8 detector and 15-keypoint skeletal pose estimation classify posture states (standing, lying, sitting, feeding).
- 3. Telemetry Streaming:** Structured behavioral records (pig ID, coordinates, posture duration) stream to the Rule Engine via MQTT/REST.
- 4. Rule Check & Anomaly Evaluation:** Engine compares metrics against temporal rules (e.g., continuous recumbency > 30 mins).
- 5. Multi-tier Alerting:** Triggers on-site strobe/siren alarm while pushing snapshot captures and incident metadata to the Web/Mobile App.

3. Hardware System Topology & Edge Interfacing

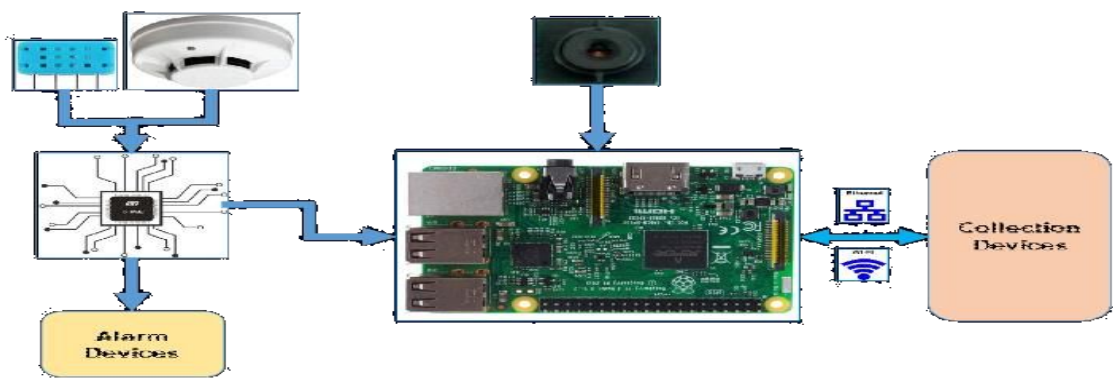


Figure 2: Modular Hardware Configuration & Sensor Interfacing Topology

Hardware Subsystem	Components & Specification	Functional Role
Edge Compute Core	Raspberry Pi 4B (4GB RAM) / Embedded Edge SBC	RTSP ingestion, lightweight YOLOv8 inference, rule engine logic
Vision Sensor	Wide-Angle IR Night Vision Camera (1080p, 120° FOV)	24/7 continuous pen coverage under zero/low ambient light
Environment Tier	ESP32 MCU + DHT22 (Temp/Hum) + MQ-135 (NH ₃ , CO ₂)	Microclimate data acquisition via I2C/UART to main edge unit
Alerting Subsystem	Relay Module + Strobe Siren + WebSocket/FCM Gateway	Instantaneous dual-channel alert (on-site visual/sound + remote push)

4. Real-time Detection Showcase & Model Validation



Figure 3A: Edge YOLO Detection (Piglet Detections: 0.89, 0.74)



Figure 3B: Multi-target Trajectory & Bounding Contour Tracking

5. Key Engineering Highlights & Practical Impact

- **Autonomous Edge Inference:** Operates completely offline with on-premise compute; requires no continuous high-bandwidth cloud video uplinks, preserving operational privacy and cutting data overhead.
- **Sub-second Anomaly Detection:** Latency from visual symptom manifestation to audio-visual alarm trigger is under 1.2 seconds, ensuring immediate intervention windows.
- **High Economic Feasibility:** Standardized bill of materials costs under per pen unit, making precision livestock monitoring accessible for small-to-medium farming cooperatives.
- **Measurable Swine Welfare Impact:** Mitigates herd mortality by identifying crushing hazards, acute lethargy, and localized asphyxiation up to 24-48 hours before conventional manual checks.